(i) Let B be the black counter

$$P(B \text{ not moved from } P \text{ to } Q) = \frac{n-k}{n}$$

$$P(B \text{ moved from } P \text{ to } Q) = \frac{k}{n}$$

Nows bag Q contains $n+k$ counters

$$P(B \text{ moved back from } Q \text{ to } P) = \frac{k}{n+k}$$

$$P(B \text{ ends in } P) = \frac{n-k}{n} + \frac{k}{n} \cdot \frac{k}{n+k} = \frac{n}{n+k}$$

$$\frac{n}{n+k} \text{ is maximised at } k = 0$$

(ii) Let B_p be the black counter in the Bag P, B_q be the black counter in the Bag Q
B_P did not move, B_Q transferred:

$$\frac{n-k}{n} \cdot \frac{k}{n+k}$$

B_P transferred to bag Q, then B_P and B_Q are transfered:

$$\frac{k}{n} \cdot \frac{k(k-1)}{(n+k)(n+k-1)}$$

Black counters end up in bag Q:

$$\frac{k}{n} \cdot \frac{n(n-1)}{(n+k)(n+k-1)}$$

$$\begin{aligned} P_k &= \frac{n-k}{n} \cdot \frac{k}{n+k} + \frac{k}{n} \cdot \frac{k(k-1)}{(n+k)(n+k-1)} + \frac{k}{n} \cdot \frac{n(n-1)}{(n+k)(n+k-1)} \\ &= \frac{2k(n-1)}{(n+k)(n+k-1)} \end{aligned}$$

$$\begin{aligned} \frac{dP_k}{dk} &= 2(n-1) \cdot \frac{(n+k)(n+k-1) - k\left((n+k) + (n+k-1)\right)}{(n+k)^2(n+k-1)^2} \\ &= 2(n-1) \cdot \frac{n(n-1) - k^2}{(n+k)^2(n+k-1)^2} \end{aligned}$$

$$\frac{dP_k}{dk} = 0 \implies k^2 = n(n-1)$$

$$k = \sqrt{n(n-1)}$$

$$k \text{ is an integer, } n-1 < \sqrt{n(n-1)} < n$$

$$P_{n-1} = \frac{2(n-1)^2}{(2n-1)(2n-2)} = \frac{n-1}{2n-1}$$

$$P_n = \frac{2n(n-1)}{(2n)(2n-1)} = \frac{n-1}{2n-1}$$

$$P_{n-1} = P_n$$

$$\therefore P_k \text{ is maximised at } k = n-1 \text{ and } k = n$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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